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TOPIC : SAFE AND GUARANTEED LEARNING-BASED CONTROL OF CYBER-PHYSICAL SYSTEMS.

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ABSTRACT : In recent years, learning-based methods have demonstrated significant success in controlling complex cyber-physical systems (CPS). These techniques involve using learning-based tools to either derive mathematical models of a system for synthesizing controllers or directly learn the controller itself. However, applying learning-based methods in safety-critical CPS poses challenges, as these components are often regarded as black-box systems without formal guarantees. In this presentation, I will explore the synergy between machine learning and control theory by addressing two key areas : firstly, how learning enhances symbolic control methods. Secondly, I will explore strategies for overcoming optimality and safety constraints in reinforcement learning algorithms through the application of control techniques.

KEYWORDS : Safety, Guaranteed Learning-based Control, Cyber-physical System.

BIOGRAPHY : Adnane SAOUD is an Assistant Professor at University Mohammed IV Polytechnic (UM6P), Benguerir, Morocco. Between September 2021 and August 2023 he was an Assistant Professor (Maitre de conférences) at CentraleSupélec, France. Between January and August 2021, he was a Postdoctoral Research Fellow in the Electrical and Computer Engineering Department at the University of California, Berkeley, USA. Between January and December 2020, he was a Postdoctoral Research Fellow in the Electrical and Computer Engineering Department at the University of California, Santa Cruz, USA. He received the Ph.D. degree in Control from CentraleSupélec, France, in 2019. In 2018, he was selected as one of the top three finalists for the Best student paper award at the European Control Conference, ECC. In 2020, he was selected as a finalist of the French national Best PhD award Discerned by the GdR MACS. In 2021, he was selected as one of the top three finalists of the EECI PhD award, recompensing the best PhD work in Europe in the field of Control. He obtained the M.Sc. degree in control from CentraleSupélec, France, in 2016, and Electrical Engineering degree from Ecole Mohammadia d'ingénieurs, (EMI), Morocco, in 2014. His current research interests include control of Cyber-Physical systems, machine learning for control and formal methods for control.



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